Transformers in Deep Learning

An Overview of Architecture, Scale, and Performance

The Transformer architecture, introduced in 2017, replaced recurrence with a self-attention mechanism that lets a model weigh the relevance of every token in a sequence against every other token simultaneously. This design enabled far greater parallelization during training than recurrent networks, and it has become the foundation for nearly all modern large language models, from BERT and GPT to multimodal systems.

This report walks through the core architecture, compares parameter scale across notable models, summarizes typical hyperparameters, and shows illustrative training behavior. Note: the training-loss curve and parameter figures below are simplified, illustrative examples meant to convey general trends rather than exact published benchmarks.

1. Core Architecture

A standard Transformer encoder block stacks a multi-head self-attention layer and a position-wise feed-forward network, each wrapped with a residual connection and layer normalization. Positional encodings are added to the input embeddings up front, since the attention mechanism itself has no inherent sense of token order.

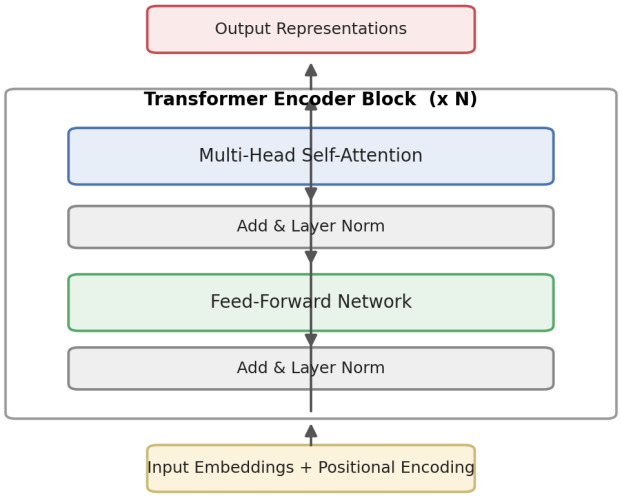


Figure 1. Simplified structure of a Transformer encoder block.

Key components:

- **Self-Attention:** computes a weighted combination of all token representations, where weights reflect learned relevance between tokens.
- **Multi-Head Attention:** runs several attention operations in parallel on different learned projections, then concatenates the results.
- **Feed-Forward Network:** a position-wise fully connected network applied identically to each token.
- **Residual Connections & Layer Norm:** stabilize training and allow deeper stacks of layers.

2. Model Comparison

Transformer-based models vary widely in scale and intended use. The table below summarizes a few widely referenced models.

Model	Year	Parameters	Layers	Primary Use Case
BERT-Base	2018	110M	12	Bidirectional language understanding
GPT-2	2019	1.5B	48	Autoregressive text generation
T5-Large	2019	770M	24	Text-to-text transfer learning
GPT-3	2020	175B	96	Few-shot general-purpose generation
PaLM	2022	540B	118	Large-scale reasoning and generation

Table 1. Approximate scale of selected Transformer models.

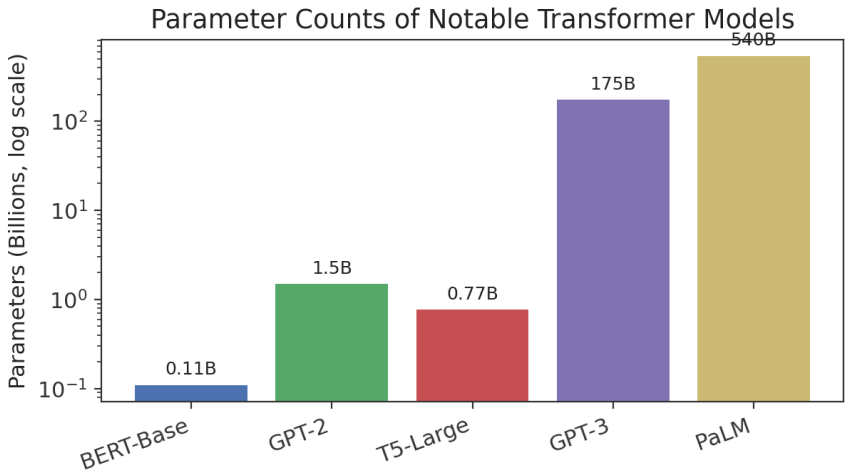


Figure 2. Parameter counts shown on a logarithmic scale, reflecting rapid growth in model size over time.

3. Training Behavior

Because self-attention allows every token to attend to every other token in a single step, Transformers typically converge faster and parallelize better than recurrent architectures such as LSTMs, which must process tokens sequentially. The chart below illustrates this general pattern with simplified example curves.

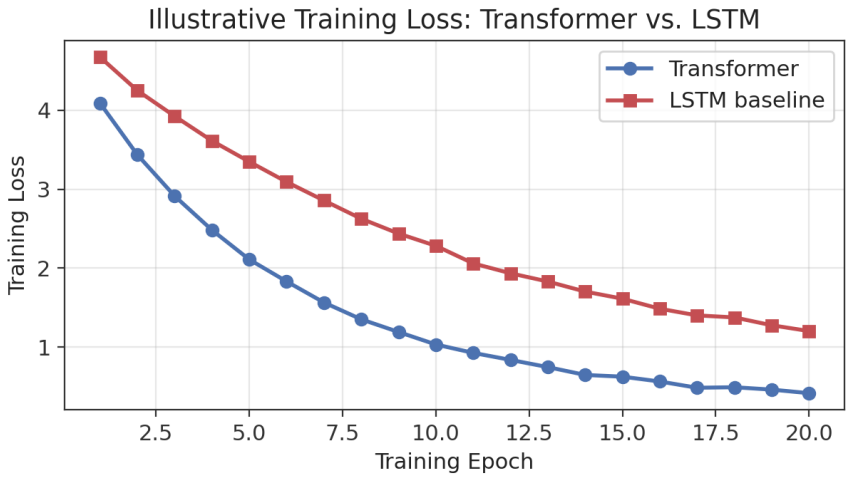


Figure 3. Illustrative training loss over 20 epochs (synthetic example data, not from a published benchmark).

Typical hyperparameters

Hyperparameter	Typical Range
Number of layers	6 - 96+
Attention heads	8 - 128
Hidden dimension	512 - 12,288
Learning rate	1e-5 - 5e-4
Warmup steps	1,000 - 10,000

Table 2. Common hyperparameter ranges reported across Transformer training setups.

4. Summary

Transformers have become the dominant architecture in deep learning largely because self-attention scales well with data and compute, parallelizes efficiently on modern hardware, and captures long-range dependencies more directly than earlier recurrent approaches. As the comparison above suggests, the field's progress has closely tracked increases in model scale, though architecture and training-technique improvements matter as well.